

YOLO Classification ONNX Model Inference Report

Backend Engine: ONNX

Phase 1: Configuration & System Metadata

Model Path	/workspace/Classification_Models/yolo26n-cls.onnx
Input Dimensions	224x224
Label Map	/workspace/cls-labels/label.txt
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/ONNX_Output
OS	Linux 6.17.0-40-generic
Python Version	3.12.3
PyTorch Version	2.13.0+cu130
Ultralytics Version	8.4.75
Inference Device	NVIDIA RTX A4000
CUDA Version	13.0
Model Architecture	YOLO (Classify)
Total Parameters	2,807,042
File Size	10.82 MB
Number of Classes	1000
Precision Mode	FP32

Model Trained Class Names:

tench, goldfish, great_white_shark, tiger_shark, hammerhead, electric_ray, stingray, cock, hen, ostrich, brambling, goldfinch, house_finch, junco, indigo_bunting, robin, bulbul, jay, magpie, chickadee, water_ouzel, kite, bald_eagle, vulture, great_grey_owl, European_fire_salamander, common_newt, eft, spotted_salamander, axolotl, bullfrog, tree_frog, tailed_frog, loggerhead, leatherback_turtle, mud_turtle, terrapin, box_turtle, banded_gecko, common_iguana, American_chameleon, whiptail, agama, frilled_lizard, alligator_lizard, Gila_monster, green_lizard, African_chameleon, Komodo_dragon, African_crocodile, American_alligator, triceratops, thunder_snake, ringneck_snake, hognose_snake, green_snake, king_snake, garter_snake, water_snake, vine_snake, night_snake, boa_constrictor, rock_python, Indian_cobra, green_mamba, sea_snake, horned_viper, diamondback, sidewinder, trilobite, harvestman, scorpion, black_and_gold_garden_spider, barn_spider, garden_spider, black_widow, tarantula, wolf_spider, tick, centipede, black_grouse, ptarmigan, ruffed_grouse, prairie_chicken, peacock, quail, partridge, African_grey, macaw, sulphur-crested_cockatoo, lorikeet, coucal, bee_eater, hornbill, hummingbird, jacamar, toucan, drake, red-breasted_merganser, goose, black_swan, tusker, echidna, platypus, wallaby, koala, wombat, jellyfish, sea_anemone, brain_coral, flatworm, nematode, conch, snail, slug, sea_slug, chiton, chambered_nautilus, Dungeness_crab, rock_crab, fiddler_crab, king_crab, American_lobster, spiny_lobster, crayfish, hermit_crab, isopod, white_stork, black_stork, spoonbill, flamingo, little_blue_heron, American_egret, bittern, crane_(bird), limpkin,

European_gallinule, American_coot, bustard, ruddy_turnstone, red-backed_sandpiper, redshank, dowitcher, oystercatcher, pelican, king_penguin, albatross, grey_whale, killer_whale, dugong, sea_lion, Chihuahua, Japanese_spaniel, Maltese_dog, Pekinese, Shih-Tzu, Blenheim_spaniel, papillon, toy_terrier, Rhodesian_ridgeback, Afghan_hound, basset, beagle, bloodhound, bluetick, black-and-tan_coonhound, Walker_hound, English_foxhound, redbone, borzoi, Irish_wolfhound, Italian_greyhound, whippet, Ibizan_hound, Norwegian_elkhound, otterhound, Saluki, Scottish_deerhound, Weimaraner, Staffordshire_bullterrier, American_Staffordshire_terrier, Bedlington_terrier, Border_terrier, Kerry_blue_terrier, Irish_terrier, Norfolk_terrier, Norwich_terrier, Yorkshire_terrier, wire-haired_fox_terrier, Lakeland_terrier, Sealyham_terrier, Airedale, cairn, Australian_terrier, Dandie_Dinmont, Boston_bull, miniature_schnauzer, giant_schnauzer, standard_schnauzer, Scotch_terrier, Tibetan_terrier, silky_terrier, soft-coated_wheaten_terrier, West_Highland_white_terrier, Lhasa, flat-coated_retriever, curly-coated_retriever, golden_retriever, Labrador_retriever, Chesapeake_Bay_retriever, German_short-haired_pointer, vizsla, English_setter, Irish_setter, Gordon_setter, Brittany_spaniel, clumber, English_springer, Welsh_springer_spaniel, cocker_spaniel, Sussex_spaniel, Irish_water_spaniel, kuvasz, schipperke, groenendael, malinois, briard, kelpie, komondor, Old_English_sheepdog, Shetland_sheepdog, collie, Border_collie, Bouvier_des_Flandres, Rottweiler, German_shepherd, Doberman, miniature_pinscher, Greater_Swiss_Mountain_dog, Bernese_mountain_dog, Appenzeller, EntleBucher, boxer, bull_mastiff, Tibetan_mastiff, French_bulldog, Great_Dane, Saint_Bernard, Eskimo_dog, malamute, Siberian_husky, dalmatian, affenpinscher, basenji, pug, Leonberg, Newfoundland, Great_Pyrenees, Samoyed, Pomeranian, chow, keeshond, Brabancon_griffon, Pembroke, Cardigan, toy_poodle, miniature_poodle, standard_poodle, Mexican_hairless, timber_wolf, white_wolf, red_wolf, coyote, dingo, dhole, African_hunting_dog, hyena, red_fox, kit_fox, Arctic_fox, grey_fox, tabby, tiger_cat, Persian_cat, Siamese_cat, Egyptian_cat, cougar, lynx, leopard, snow_leopard, jaguar, lion, tiger, cheetah, brown_bear, American_black_bear, ice_bear, sloth_bear, mongoose, meerkat, tiger_beetle, ladybug, ground_beetle, long-horned_beetle, leaf_beetle, dung_beetle, rhinoceros_beetle, weevil, fly, bee, ant, grasshopper, cricket, walking_stick, cockroach, mantis, cicada, leafhopper, lacewing, dragonfly, damselfly, admiral, ringlet, monarch, cabbage_butterfly, sulphur_butterfly, lycaenid, starfish, sea_urchin, sea_cucumber, wood_rabbit, hare, Angora, hamster, porcupine, fox_squirrel, marmot, beaver, guinea_pig, sorrel, zebra, hog, wild_boar, warthog, hippopotamus, ox, water_buffalo, bison, ram, bighorn, ibex, hartebeest, impala, gazelle, Arabian_camel, llama, weasel, mink, polecat, black-footed_ferret, otter, skunk, badger, armadillo, three-toed_sloth, orangutan, gorilla, chimpanzee, gibbon, siamang, guenon, patas, baboon, macaque, langur, colobus, proboscis_monkey, marmoset, capuchin, howler_monkey, titi, spider_monkey, squirrel_monkey, Madagascar_cat, indri, Indian_elephant, African_elephant, lesser_panda, giant_panda, barracouta, eel, coho, rock_beauty, anemone_fish, sturgeon, gar, lionfish, puffer, abacus, abaya, academic_gown, accordion, acoustic_guitar, aircraft_carrier, airliner, airship, altar, ambulance, amphibian, analog_clock, apiary, apron, ashcan, assault_rifle, backpack, bakery, balance_beam, balloon, ballpoint, Band_Aid, banjo, bannister, barbell, barber_chair, barbershop, barn, barometer, barrel, barrow, baseball, basketball, bassinet, bassoon, bathing_cap, bath_towel, bathtub, beach_wagon, beacon, beaker, bearskin, beer_bottle, beer_glass, bell_cote, bib, bicycle-built-for-two, bikini, binder, binoculars, birdhouse, boathouse, bobsled, bolo_tie, bonnet, bookcase, bookshop, bottlecap, bow, bow_tie, brass, brassiere, breakwater, breastplate, broom, bucket, buckle, bulletproof_vest, bullet_train, butcher_shop, cab, caldron, candle, cannon, canoe, can_opener, cardigan, car_mirror, carousel, carpenter's_kit, carton, car_wheel, cash_machine, cassette, cassette_player, castle, catamaran, CD_player, cello, cellular_telephone, chain, chainlink_fence, chain_mail, chain_saw, chest, chiffonier, chime, china_cabinet, Christmas_stocking, church, cinema, cleaver, cliff_dwelling, cloak, clog, cocktail_shaker, coffee_mug, coffeepot, coil, combination_lock, computer_keyboard, confectionery, container_ship, convertible, corkscrew, cornet, cowboy_boot, cowboy_hat, cradle, crane_(machine), crash_helmet, crate, crib, Crock_Pot, croquet_ball, crutch, cuirass, dam, desk, desktop_computer, dial_telephone, diaper, digital_clock, digital_watch, dining_table, dishrag, dishwasher, disk_brake, dock, dogsled, dome, doormat, drilling_platform, drum, drumstick, dumbbell, Dutch_oven, electric_fan, electric_guitar, electric_locomotive, entertainment_center, envelope, espresso_maker, face_powder, feather_boa, file, fireboat, fire_engine, fire_screen, flagpole, flute, folding_chair, football_helmet, forklift, fountain, fountain_pen, four-poster, freight_car, French_horn, frying_pan, fur_coat, garbage_truck, gasmask, gas_pump, goblet, go-kart, golf_ball, golfcart, gondola, gong, gown, grand_piano, greenhouse, grille, grocery_store, guillotine, hair_slide, hair_spray, half_track, hammer, hamper, hand_blower,

hand-held_computer, handkerchief, hard_disc, harmonica, harp, harvester, hatchet, holster, home_theater, honeycomb, hook, hoopskirt, horizontal_bar, horse_cart, hourglass, iPod, iron, jack-o'-lantern, jean, jeep, jersey, jigsaw_puzzle, jinrikisha, joystick, kimono, knee_pad, knot, lab_coat, ladle, lampshade, laptop, lawn_mower, lens_cap, letter_opener, library, lifeboat, lighter, limousine, liner, lipstick, Loafer, lotion, loudspeaker, loupe, lumbermill, magnetic_compass, mailbag, mailbox, maillot_(tights), maillot_(tank_suit), manhole_cover, maraca, marimba, mask, matchstick, maypole, maze, measuring_cup, medicine_chest, megalith, microphone, microwave, military_uniform, milk_can, minibus, miniskirt, minivan, missile, mitten, mixing_bowl, mobile_home, Model_T, modem, monastery, monitor, moped, mortar, mortarboard, mosque, mosquito_net, motor_scooter, mountain_bike, mountain_tent, mouse, mousetrap, moving_van, muzzle, nail, neck_brace, necklace, nipple, notebook, obelisk, oboe, ocarina, odometer, oil_filter, organ, oscilloscope, overskirt, oxcart, oxygen_mask, packet, paddle, paddlewheel, padlock, paintbrush, pajama, palace, panpipe, paper_towel, parachute, parallel_bars, park_bench, parking_meter, passenger_car, patio, pay-phone, pedestal, pencil_box, pencil_sharpener, perfume, Petri_dish, photocopier, pick, pickelhaube, picket_fence, pickup, pier, piggy_bank, pill_bottle, pillow, ping-pong_ball, pinwheel, pirate, pitcher, plane, planetarium, plastic_bag, plate Rack, plow, plunger, Polaroid_camera, pole, police_van, poncho, pool_table, pop_bottle, pot, potter's_wheel, power_drill, prayer_rug, printer, prison, projectile, projector, puck, punching_bag, purse, quill, quilt, racer, racket, radiator, radio, radio_telescope, rain_barrel, recreational_vehicle, reel, reflex_camera, refrigerator, remote_control, restaurant, revolver, rifle, rocking_chair, rotisserie, rubber_eraser, rugby_ball, rule, running_shoe, safe, safety_pin, saltshaker, sandal, sarong, sax, scabbard, scale, school_bus, schooner, scoreboard, screen, screw, screwdriver, seat_belt, sewing_machine, shield, shoe_shop, shoji, shopping_basket, shopping_cart, shovel, shower_cap, shower_curtain, ski, ski_mask, sleeping_bag, slide_rule, sliding_door, slot, snorkel, snowmobile, snowplow, soap_dispenser, soccer_ball, sock, solar_dish, sombrero, soup_bowl, space_bar, space_heater, space_shuttle, spatula, speedboat, spider_web, spindle, sports_car, spotlight, stage, steam_locomotive, steel_arch_bridge, steel_drum, stethoscope, stole, stone_wall, stopwatch, stove, strainer, streetcar, stretcher, studio_couch, stupa, submarine, suit, sundial, sunglass, sunglasses, sunscreen, suspension_bridge, swab, sweatshirt, swimming_trunks, swing, switch, syringe, table_lamp, tank, tape_player, teapot, teddy, television, tennis_ball, thatch, theater_curtain, thimble, thresher, throne, tile_roof, toaster, tobacco_shop, toilet_seat, torch, totem_pole, tow_truck, toyshop, tractor, trailer_truck, tray, trench_coat, tricycle, trimaran, tripod, triumphal_arch, trolleybus, trombone, tub, turnstile, typewriter_keyboard, umbrella, unicycle, upright, vacuum, vase, vault, velvet, vending_machine, vestment, viaduct, violin, volleyball, waffle_iron, wall_clock, wallet, wardrobe, warplane, washbasin, washer, water_bottle, water_jug, water_tower, whiskey_jug, whistle, wig, window_screen, window_shade, Windsor_tie, wine_bottle, wing, wok, wooden_spoon, wool, worm_fence, wreck, yawl, yurt, web_site, comic_book, crossword_puzzle, street_sign, traffic_light, book_jacket, menu, plate, guacamole, consomme, hot_pot, trifle, ice_cream, ice_lolly, French_loaf, bagel, pretzel, cheeseburger, hotdog, mashed_potato, head_cabbage, broccoli, cauliflower, zucchini, spaghetti_squash, acorn_squash, butternut_squash, cucumber, artichoke, bell_pepper, cardoon, mushroom, Granny_Smith, strawberry, orange, lemon, fig, pineapple, banana, jackfruit, custard_apple, pomegranate, hay, carbonara, chocolate_sauce, dough, meat_loaf, pizza, potpie, burrito, red_wine, espresso, cup, eggnog, alp, bubble, cliff, coral_reef, geyser, lakeside, promontory, sandbar, seashore, valley, volcano, ballplayer, groom, scuba_diver, rapeseed, daisy, yellow_lady's_slipper, corn, acorn, hip, buckeye, coral_fungus, agaric, gyromitra, stinkhorn, earthstar, hen-of-the-woods, bolete, ear, toilet_tissue

Phase 2: Inference Performance Metrics

Total Images Processed	5
Total Pipeline Time	1043.84 ms
Preprocess Latency	56.55 ms per image
Inference Latency	43.72 ms per image
Postprocess Latency	108.50 ms per image
Throughput	4.79 Images/sec
Peak Memory Usage	8.50 MB

Phase 3: Detailed Predictions Summary

Image File	Predicted Class	Confidence
Cat_November_2010-1a.jpg.webp	tiger cat	39.4%
images.jpeg	Saluki	79.2%
images1.jpeg	tiger	75.5%
istockphoto-1189925691-612x612.jpg	passenger car	73.7%
tigermain.jpg	tiger	94.1%